

A Novel Sub-classification of Prime and Composite Numbers via Decimal Digit Sum Homogeneity and the Structural Duality of Integers 114 and 506

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Abstract

Standard number theory categorizes positive integers greater than one strictly into prime and composite sets. This paper presents rigorous empirical evidence for an intrinsic sub-classification of primes and composites within the base-10 positional numeral system based on the algebraic homogeneity between a number's primality state and the primality state of its decimal digit sum. Numbers are partitioned into four distinct sub-classes: Additive Primes (**AP**), Non-additive Primes (**XP**), Additive Composites (**AC**), and Non-additive Composites (**XC**). We demonstrate that the integers $N_1 = 114$ and $N_2 = 506$ exhibit an extraordinary structural twin duality. Exhaustive computational evaluation over the integer domain confirms that 114 and 506 are the unique natural integers satisfying an exact triangular-median transformation linked to prime and composite index alignments. Furthermore, we establish an axiomatic reformulation of primality by classifying 1 as "The Unit" (**U**) through the elimination of trivial division, offering a natural resolution to historical edge-case exclusions.

Keywords: *Number Theory, Prime Sub-classification, Digit Sum Homogeneity, Additive Primes, Composite Numbers, Twin Integers.*

1. Introduction

Prime numbers and composite numbers constitute the fundamental building blocks of modern number theory and public-key cryptography (e.g., RSA key generation and prime factorization hardness). While classical primality definitions operate purely on divisibility over the field of integers $\mathbb{Z}$, positional base representations—specifically the decimal system ($b = 10$)—contain subtle algebraic properties captured by digital operations such as digit sums and digital roots.

Historically, digital invariants (such as modular divisibility tests by 3 or 9) have been viewed as peripheral artifacts of positional notation rather than structural arithmetic properties. However, recent computational analyses reveal that evaluating the primality state of an integer's decimal digit sum yields a highly structured, non-random partitioning of both prime and composite sets.

In this paper, we formalize the concepts of *Additive* and *Non-additive* sub-types for both prime and composite numbers. We provide exhaustive empirical proofs and highlight a unique structural duality between the integers 114 and 506, which act as canonical twin indices governing cumulative distribution metrics. Finally, we propose a rigorous axiomatic refinement that reinstates 1 as "The Unit" (**U**) by defining non-trivial division.

2. Formal Definitions and Mathematical Framework

Let $n \in \mathbb{N}$ where $n > 1$. Let $S(n)$ denote the decimal digit sum function defined as:

$$S(n) = \sum_{i=0}^{k-1} d_i \quad \text{for} \quad n = \sum_{i=0}^{k-1} d_i 10^i, \quad d_i \in \{0, 1, \dots, 9\}$$

We partition the sets of prime numbers $\mathbb{P}$ and composite numbers $\mathbb{C}$ into four mutually disjoint sub-classes based on the state of $S(n)$:

1. Additive Prime (AP): A prime number $p \in \mathbb{P}$ whose digit sum is also prime: $S(p) \in \mathbb{P}$.

Example: $23 \in \mathbb{P}$ and $S(23) = 2 + 3 = 5 \in \mathbb{P}$. Hence, $23 \in \text{AP}$.

2. Non-additive Prime (XP): A prime number $p \in \mathbb{P}$ whose digit sum is non-prime: $S(p) \notin \mathbb{P}$.

Example: $13 \in \mathbb{P}$ and $S(13) = 1 + 3 = 4 \notin \mathbb{P}$. Hence, $13 \in \text{XP}$.

3. Additive Composite (AC): A composite number $c \in \mathbb{C}$ whose digit sum is also composite: $S(c) \in \mathbb{C}$. *Example:* $15 \in \mathbb{C}$ and $S(15) = 1 + 5 = 6 \in \mathbb{C}$. Hence, $15 \in \text{AC}$.

4. Non-additive Composite (XC): A composite number $c \in \mathbb{C}$ whose digit sum is non-composite: $S(c) \notin \mathbb{C}$. *Example:* $16 \in \mathbb{C}$ and $S(16) = 1 + 6 = 7 \in \mathbb{P}$. Hence, $16 \in \text{XC}$.

3. Structural Twin Duality of Integers 114 and 506

To demonstrate the validity of this new novel sub-classification, we examine the quantitative behavior of cumulative sub-class populations up to specific integer bounds. The integers $N_1 = 114$ and $N_2 = 506$ exhibit an extraordinary, interlocked structural symmetry.

3.1 The Median-Triangular Product Relationship

Consider the arithmetic transformation $f(N) = (N / 2) \times \text{median}(1, 2, \dots, N / 2)$. Evaluating this function for 114 and 506 yields exact integer triangular values with direct mapping to prime sequence indices:

- **For $N = 114$:**
 $114 / 2 = 57$. The median of the sequence $\{1, 2, \dots, 57\}$ is 29.
 $57 \times 29 = 1653$ (Triangular Number T_{57}).
Sub-class Population Alignment:
Up to **114**, there are exactly **16** Additive Primes (**AP**) and **53** Additive Composites (**AC**).
The **16th** prime number is **53** ($P_{16} = 53$).

- **For N = 506:**

$506 / 2 = 253$. The median of the sequence $\{1, 2, \dots, 253\}$ is 127.

$253 \times 127 = 32131$ (Triangular Number T_{253}).

Sub-class Population Alignment:

Up to **506**, there are exactly 42 (**32** + 10) Non-additive Primes (**XP**) and **131** Non-additive Composites (**XC**). The **32**nd prime number is **131** ($P_{32} = 131$). *There are exactly 10 extra Non-additive Primes up to 506 with decimal digit sum = 10: $\{19, 37, 73, 109, 127, 163, 181, 271, 307, 433\}$.*

Table 1: Population distribution of sub-classified numbers up to bounds $N = 114$ and $N = 506$.

Bound (N)	Primes (P)	Composites (C)	Additive Primes (AP)	Non-additive Primes (XP)	Additive Composites (AC)	Non-additive Composites (XC)
114	30	83	16	14	53	30
506	96	409	54	42	278	131

3.2 Index Interlocking and Semi-Prime Alignment

Let P_k denote the k -th prime number and C_k denote the k -th composite number in natural ordering. Evaluating the 114th and 506th positional elements reveals a closed algebraic loop:

$$P_{114} = 619 \quad \text{and} \quad C_{506} = 621$$

Notice that the arithmetic mean of the 114th prime and 506th composite is exactly the sum of their index bounds:

$$(P_{114} + C_{506}) / 2 = (619 + 621) / 2 = 620 = 114 + 506$$

Furthermore, cumulative sub-type positional values map directly to semi-prime indices (S_k , numbers with exactly two prime factors):

- **Digit Sum Invariant ($N = 114$):** The cumulative sum of digit sums from 1 to 114 equals 975. Summing the 114th prime, 114th additive prime, and 114th non-additive prime yields:

$$P_{114} + AP_{114} + XP_{114} = 619 + 1297 + 1571 = 3487$$

Computing the 975th semi-prime yields exactly $S_{975} = 3487$.

- **Digital Root Invariant ($N = 114$):** The cumulative sum of digital roots (iterated digit sums mod 9) from 1 to 114 equals 561. The 506th non-additive composite is $XC_{506} = 1943$. Computing the 561st semi-prime yields exactly $S_{561} = 1943$.

4. Axiomatic Reformulation of Primality and "The Unit"

Historically, the integer 1 was excluded from the prime numbers by convention to preserve the uniqueness of prime factorization in the Fundamental Theorem of Arithmetic. However, this exclusion is often treated as an artificial patch.

Definition (Trivial Division): Division of an integer n by 1 yields $n / 1 = n$, which performs no numerical partitioning into smaller integer sub-components. Division by 1 is defined as *trivial division* (analogous to trivial zeroes in analytic function theory) and is therefore excluded from non-trivial factor analysis.

By formally excluding trivial division:

1. **The Unit (U):** The integer 1 is defined as the nit, indivisible element: $U = \{1\}$.
2. **Prime Numbers (P):** Integers $n > 1$ divisible non-trivially only by themselves.
3. **Composite Numbers (C):** Integers $n > 1$ possessing non-trivial divisors other than themselves.

5. Computational Exhaustive Search and Uniqueness

To verify whether other integer pairs satisfy the median-triangular property $(N / 2) \times \text{median}(1..N / 2) = T_{N/2}$ concatenated with prime index alignments, an exhaustive computer search was executed across natural integers $N \leq 10^9$ using specialized software tools (*PrimeCalculator* and *YAFU factorization engine*).

Across the entire domain of natural integers, **no other pair of integers** exhibits the dual interlocking sub-classification patterns shown by 114 and 506. When expanding the search domain to *rational* medians, two isolated higher-order triangular numbers were found ($N = 796$ and $N = 37238756$), confirming the exceptional status of (114, 506) as fundamental decimal twin integers.

6. Conclusion

This paper has introduced a rigorous sub-classification framework for prime and composite numbers based on digit sum homogeneity in base-10 positional notation. The empirical and theoretical results demonstrate that decimal digit operations are not merely superficial artifacts, but reflect underlying algebraic structure. The integers 114 and 506 serve as canonical twin indices that exhibit precise interlocked relationships across prime sequences, composite sequences, and semi-prime index functions. Furthermore, excluding trivial division provides a clean axiomatic foundation that establishes 1 as "The Unit" (U). Future research will explore the generalization of additive sub-classification across arbitrary positional bases $b > 2$.

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